

v0.20.1 Workday Visual + Interaction Repair

This pass upgrades the Attention, MLP, Layers, and Hidden States Journey cards with richer lesson fields, clearer coded visuals, dedicated tiny interactions, and glossary support. It keeps Journey order, progress, checkpoint randomization, the one-badge model, games, dependencies, and generated assets unchanged.

What Changed

Attention

Concrete token relevance replaces abstract token nodes. The interaction asks which earlier token helps interpret "it."

MLP

Feature bars show the same token position before and after per-token MLP reshaping.

Layers

A repeated attention-plus-MLP stack shows refinement without implying human thought steps.

Hidden States

A state-flow diagram and sort interaction distinguish durable weights from temporary activations.

Files Changed

- `src/data/content.ts`
- `src/data/exercises.ts`
- `src/components/VisualAids.tsx`
- `src/main.tsx`
- `src/styles/global.css`
- `docs/REVIEW_NOTES.md`
- `docs/stage-audits/v0-20-workday/*` implementation docs, screenshots, and this report

Glossary Updates

Added or improved Attention, Relevance weight, MLP, Multi-layer perceptron, Feed-forward network, Layer, Transformer layer, Residual path, Normalization, Hidden state, and Temporary activation.

Screenshots

3

Workday

Transformer layers process the current context.

The prompt is inside the model, being reshaped by repeated transformer layers.

FOCUS

How transformer layers process the current context.

KEEP CLEAR

Attention is relevance, not awareness.

16

Attention

Weighted relevance between positions

Preview

17

MLP

Feature reshaping per token

Preview

18

Layers

Repeated refinement

Preview



Hidden States



Home



Journey



Play



Glossary



Badge

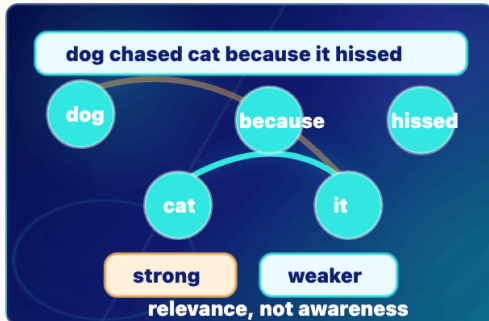
Workday overview at 390px.

context.

Previewing this card. Progress will not change.

VISUAL AID

What to picture



Attention Weave

Relevance, not awareness

Attention assigns temporary relevance weights between token positions in the current context.

- 1 Current context** The visible tokens include dog, cat, because, it, and hissed.
- 2 Strong relation** In this sentence, it is most relevant to cat.
- 3 Weaker relation** dog is nearby and possible in other sentences, but it hissed points to cat here.
- 4 Limit** Attention computes relevance



Home



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Glossary



Badge

Attention visual: relevance between token positions.

TINY INTERACTION

Connect the pronoun

Which earlier token helps interpret "it"?

dog cat because it hissed

dog

cat

Insight strengthened. Attention can weight "cat" as more relevant to "it" in this context.

Choose which earlier token helps interpret "it." Attention is relevance, not awareness.

CHECKPOINT

What does attention do in a transformer?

A It chooses the final answer by itself.

B It permanently stores memories.



Home



Journey



Play

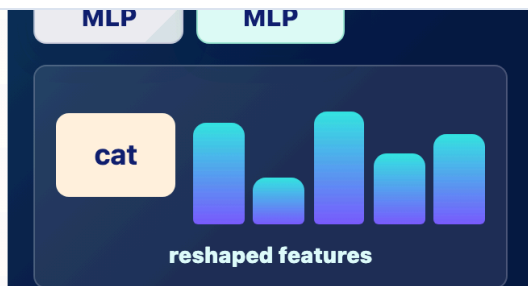


Glossary



Badge

Attention tiny interaction feedback.



What changed?

The token's feature vector

The model's permanent memory

The token ID

Insight strengthened. The MLP reshapes features for a token position during the forward pass.

Toggle the same token position before and after the MLP. The feature vector changes; the token ID and weights do not.

CHECKPOINT

What does the MLP mainly do here?

- A Mixes information between different token positions
- B Reshapes each token position's feature vector

Home Journey Play Glossary Badge

MLP feature toggle interaction.



Insight strengthened. The refined hidden-state ribbon moves forward and is transformed again. Hidden states are refined, not permanently stored.

Tap a layer to watch the same hidden-state ribbon move forward through repeated attention and MLP blocks.

CHECKPOINT

What do layers repeatedly refine?

- A The training dataset
- B Permanent user memory

Home Journey Play Glossary Badge

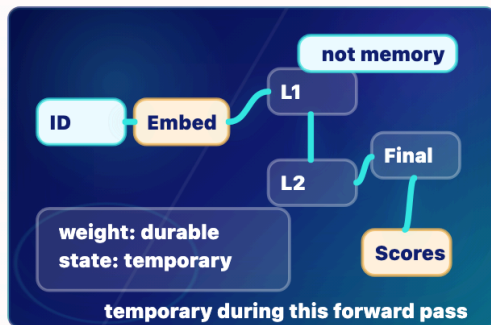
Layers stack inspector.

a particular layer.

Previewing this card. Progress will not change.

VISUAL AID

What to picture



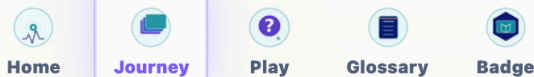
Hidden State Flow

Temporary during this forward pass

A token starts with an embedding and becomes context-shaped hidden states as it moves through layers.

- 1 Embedding** A learned starting vector is retrieved for a token ID.
- 2 Hidden states** Layers reshape that vector into temporary context-shaped states.
- 3 Scores later** Final hidden states help produce next-token scores.
- 4 Not memory** A hidden state is not visible text, permanent memory, or a stored fact.

Key takeaway:



Hidden States flow visual.

hidden state Temporary

attention pattern Temporary

current context Temporary

training data Not in this forward pass

Durable

Temporary

Not in this forward pass

Insight strengthened. Hidden states are temporary internal vectors shaped by the current context.

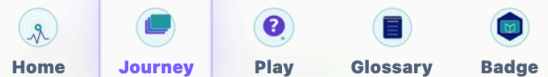
Sort model parts by whether they persist as learned parameters, appear temporarily during this run, or sit outside the current forward pass.

CHECKPOINT

Why is it called hidden state?

A It is internal numbers, not visible text

B It is a secret memory file



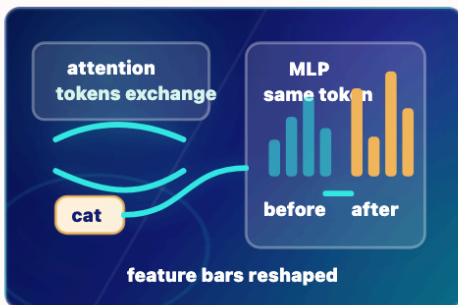
Hidden States sort interaction.

forward network that
reshapes each token's
feature vector inside a
transformer layer.

Previewing this card.
Progress will not change.

VISUAL AID

What to picture



MLP Feature Workshop

Same token, reshaped features

After attention mixes across tokens, the MLP reshapes features within each token position.

- 1 **Attention first** Token positions can exchange current-context information.
- 2 **Same position** The example token cat stays in the same position.

3 **MLP reshapes Feature here**



Home



Journey



Play



Glossary



Badge

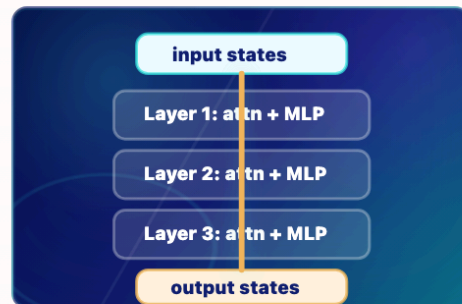
MLP visual at 320px.

through attention, MLPs,
residual paths, and
normalization.

Previewing this card.
Progress will not change.

VISUAL AID

What to picture



Transformer Stack

Repeated attention plus MLP

Layers repeat attention and MLP transformations to refine temporary hidden states.

- 1 **Input states** Current-context hidden states enter the stack.
- 2 **Repeated block** Each layer applies attention and an MLP, with residual paths and normalization helping signal flow.
- 3 **Output states** Refined hidden states continue toward next-token scores.



Home



Journey



Play



Glossary



Badge

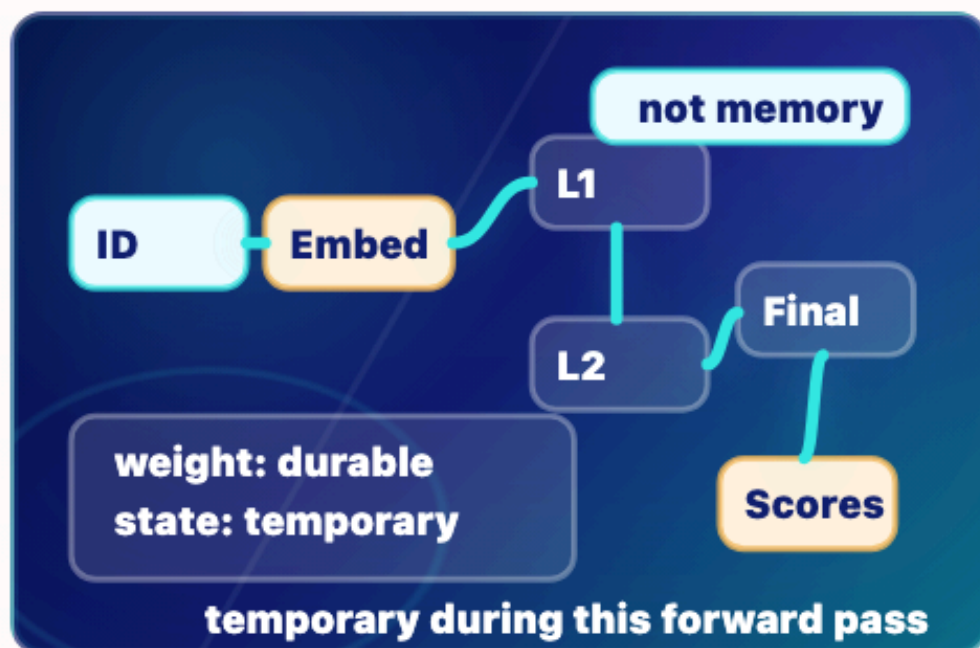
Layers visual at 320px.

**temporary context-shaped
vector for a token at a
particular layer.**

**Previewing this card.
Progress will not change.**

VISUAL AID

What to picture



Hidden State Flow

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Home



Journey



Play



Glossary



Badge

Hidden States visual at 320px.

Verification

- `npm run typecheck`: **passed**
- `npm run build`: **passed** with the existing Vite large-chunk warning
- `npm run build:pages`: **passed** with the existing Vite large-chunk warning

- `npm run audit:answers:` **passed**
- Browser QA: **passed** for captured 390px and 320px Workday states; no horizontal overflow was detected.

Known Issues

- Long cards still require scrolling, and the bottom nav can appear at the bottom of tall screenshots. Required controls remain reachable.
- The worktree also contains pre-existing uncommitted v0.19.3 Glossary Dojo feedback files that were not part of this pass.
- Workday is not yet a strong Image 2 target. Coded diagrams remain the right teaching surface here.